미적분2_3차과제

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11.5

1. $v = \langle 4-1, -1-1, 2-5 \rangle = \langle -3, +2, +3 \rangle \rightarrow x = -3t+4,\ y = +2t-1,\ z = 2+3t$

$\frac{x-4}{-3} = \frac{y+1}{2} = \frac{z-2}{3}$

2. $v = \langle 3, 2, 1 \rangle$ $x = 3t+1,\ y = 2t,\ z = t-1$

3. $n = \langle 2, -1, 5 \rangle = V$ $x = 2t-2,\ y = -t+2,\ z = 5t+4$ $\frac{x+2}{2} = 2-y = \frac{z-4}{5}$

4. $L_1\ V = \langle 2, -2, 7 \rangle$ $L_2\ V = \langle 1, -1, 3 \rangle$ 평행 X

$2t+2 = 5$, $-2t+3 = -5s+1$, $7t = 3s+2$ 꼬인위치

$5 = 1$

5. (1) $n = \langle 1, 4, -3 \rangle \rightarrow (x-2) + 4(y-1) - 3(z-0) = 0 \rightarrow x + 4y - 3z = 6$

(2) $a = \langle 1, 1, 1 \rangle$ $b = \langle 3, 4, 0 \rangle$ $n = a \times b = \langle -4, 3, 1 \rangle \rightarrow -4(x-3) + 3(y+1) + (z-1) = 0$

$\rightarrow -4x + 3y + z = -14$

6. $n = \langle 1, 1, -1 \rangle$, $n_2 = \langle 2, -3, 4 \rangle$ 평행 X $n \cdot n_2 = 2 - 3 - 4 = -5$ 수직 X

$\cos\theta = \frac{|n_1 \cdot n_2|}{|n_1||n_2|} = \frac{5}{\sqrt{3}\sqrt{29}} = \frac{5}{\sqrt{87}}$ $\theta = \cos^{-1}\left(\frac{5}{\sqrt{87}}\right)$

7. $x + 3y - z - 2 = 0$ $(0, 0, 4)$ $\frac{4-2}{\sqrt{11}} = \frac{2}{\sqrt{11}}$

8. (1) $\vec{AB} = \langle 2, -4, 4 \rangle$, $\vec{AC} = \langle 3, -1, -1 \rangle$ $n = \vec{AB} \times \vec{AC} = \langle 8, 14, 10 \rangle$

$8(x-2) + 14(y-3) + 10(z-1) = 0$ $4x + 7y + 5z - 34 = 0$

(2) $S = \frac{1}{2}|\vec{AB} \times \vec{AC}| = \frac{1}{2}\sqrt{8^2 + 14^2 + 10^2} = 3\sqrt{10}$

(3) $\frac{|-34+1|}{\sqrt{4^2+7^2+5^2}} = \frac{34}{3\sqrt{10}} \times 3\sqrt{10} \times \frac{1}{3} = \frac{34}{3}$

12.1

1. (a) $\sqrt{2-t}$의 정의역 $= \{t \le 2\}$, $\frac{e^t-1}{t}$의 정의역 $= \{t \neq 0\}$, $\ln(t+1)$의 정의역 $= \{t > -1\}$

$r(t)$의 정의역 $= \{-1 < t < 0,\ 0 < t \le 2\}$

(b) $\lim_{t \to 0} r(t) = \langle \sqrt{2}, 1, 0 \rangle$

2. $r(t) = (1-t)\langle 1, 2, 3 \rangle + t\langle 4, 5, 6 \rangle = \langle 1-t+4t, 2-2t+5t, 3-3t+6t \rangle$

$= \langle 1+3t, 2+3t, 3+3t \rangle$, $0 \le t \le 1$, $x = 1+3t$, $y = 2+3t$, $z = 3+3t$ $0 \le t \le 1$

3. $x^2 + y^2 = 16 = r^2$ $r = 4$ $z = 5 - x$ $x = 4\cos t,\ y = 4\sin t,\ z = 5 - 4\cos t$

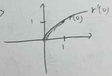
$\rightarrow r(t) = \langle 4\cos t, 4\sin t, 5 - 4\cos t \rangle$, $0 \le t \le 2\pi$

12.2

1. $r'(t) = \frac{1}{1+t^2} i + 4e^{4t} j + (8te^t + 8e^t) k = \langle \frac{1}{1+t^2}, 4e^{4t}, 8(t+1)e^t \rangle$

(b) $r'(0) = \langle 1, 4, 8 \rangle$, $|r'(0)| = 9$ $T(0) = \frac{\langle 1, 4, 8 \rangle}{9} = \langle \frac{1}{9}, \frac{4}{9}, \frac{8}{9} \rangle$

2. $r'(t) = \langle 2e^{2t}, e^t \rangle$ $r'(0) = \langle 2, 1 \rangle$, $r(0) = \langle 1, 1 \rangle$



3. $r(t) = \langle t\cos t, t, t\sin t \rangle$ $r'(t) = \langle \cos t - t\sin t, 1, \sin t + t\cos t \rangle$

$r'(\pi) = \langle -1, 1, -\pi \rangle$ $\frac{x+\pi}{-1} = \frac{y-\pi}{1} = \frac{z}{-\pi}$

4. $n = \langle \sqrt{3}, 1, 0 \rangle$ $r'(t) = \langle -2\sin t, 2\cos t, e^t \rangle$ $n \cdot r'(t) = -2\sqrt{3}\sin t + 2\cos t = 0$

$\tan = \frac{1}{\sqrt{3}} \rightarrow t = \frac{\pi}{6}$ $r(\frac{\pi}{6}) = \langle \sqrt{3}, 1, e^{\frac{\pi}{6}} \rangle$ $r'(\frac{\pi}{6}) = \langle -1, \sqrt{3}, e^{\frac{\pi}{6}} \rangle$

$\frac{x-\sqrt{3}}{-1}, \frac{y-1}{\sqrt{3}} = \frac{z - e^{\frac{\pi}{6}}}{e^{\frac{\pi}{6}}}$

5. $\left[\langle \frac{1}{3}t^3 + C_1, \frac{t}{\pi}\sin\pi t + \frac{1}{\pi^2}\cos\pi t + C_2, -\frac{1}{\pi}\cos\pi t + C_3 \rangle \right]_0^1 = \langle \frac{1}{3}, -\frac{2}{\pi^2}, \frac{2}{\pi} \rangle$

r(t)
1. |r'(t)|
12.4
1. r(t
a
2. a
V
r

12.3

$r(t) = \langle 2e^t\cos t, 2e^t\sin t, 4e^t \rangle$, $r'(t) = \langle 2e^t(\cos t - \sin t), 2e^t(\sin t + \cos t), 4e^t \rangle$

$|r'(t)| = 2e^t\sqrt{(\cos t - \sin t)^2 + (\sin t + \cos t)^2 + 4} = 2\sqrt{6}e^t$ $\int_0^1 2\sqrt{6}e^t\,dt = 2\sqrt{6}(e-1)$

12.4

$r(t) = \langle t\ln t, t, e^{-t} \rangle$ $v = r'(t) = \langle \ln t + 1, 1, -e^{-t} \rangle$ $|v| = \sqrt{(\ln t + 1)^2 + 1^2 + e^{-2t}}$

$a = r''(t) = \langle \frac{1}{t}, 0, e^{-t} \rangle$

$a(t) = \langle 6t, 12t^2, -6t \rangle \to v(t) = \langle 3t^2 + C_1, 4t^3 + C_2, -3t^2 + C_3 \rangle$

$v(0) = \langle 1, -1, 3 \rangle = \langle C_1, C_2, C_3 \rangle \to v(t) = \langle 3t^2 + 1, 4t^3 - 1, -3t^2 + 3 \rangle$

$r(t) = \langle t^3 + t + d_1, t^4 - t + d_2, -t^3 + 3t + d \rangle \to \langle 0, 0, 0 \rangle = r(0) = \langle d_1, d_2, d_3 \rangle$

$\to r(t) = \langle t^3 + t, t^4 - t, -t^3 + 3t \rangle$